

Firas Adnan Jalil

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EDUCATION

University of Toronto

Sept 2024 – May 2028

HBSc Computer Science, Arts & Science Internship Program

Toronto, ON

- GPA: 3.56
- Courses: Software Design, Software Tools & System Programming, Linear Algebra I, Data Structures & Analysis

SUMMARY OF QUALIFICATIONS

Languages: Python, JavaScript, C, TypeScript, Java

Tools: React, NextJS, TailwindCSS, FastAPI, Flask, Flutter, Celery, Redis, Firebase, Supabase, PostgreSQL

Machine Learning: TensorFlow, Scikit-Learn, Pandas, NumPy, Matplotlib

Developer Tools: Git, Figma, Docker

EXPERIENCE

Systems Engineer

Sept 2025 – Apr 2026

Muslims in Canada Archives

Toronto, ON

- Visualized system architecture by mapping frontend components to backend rendering logic on **Miro**, eliminating ambiguity to streamline complex debugging analysis
- Reduced average technical onboarding time for new developers by **25%** by comprehensively documenting the full data rendering flow across the CMS, codebase and database
- Traced **Typesense** search service usage throughout codebase to provide recommendations on a caching algorithm implementation, reducing average user-facing query times by **20%**

Data Analyst Intern

Sept 2024 – Jan 2025

Vertige Investment Group

Toronto, ON

- Developed an ETL pipeline to automate the processing and standardization of financial data, leveraging **Scikit-Learn** and **Numpy** for enriching data from Yahoo Finance API
- Architected a multi-variate LSTM neural network to process multi-ticker feature sets, utilizing ticker-specific normalization and one-hot encoding to isolate and scale time-series data for 8+ financial assets
- Engineered a rigorous backtest framework to assess strategy viability, quantifying risk using Sharpe ratio and maximum drawdown

PROJECTS

Syncer | *React, FastAPI, Celery, Redis, Supabase*

- Launched a Chrome extension to sync Spotify/YouTube playlists, building a responsive interface with **React/TailwindCSS** and a **FastAPI** backend that uses **Celery + Redis** to manage long-running tasks
- Implemented cross-platform song matching by tokenizing song metadata and fuzzy-matching results using a custom Levenshtein algorithm, achieving ~95% match rate on test playlists
- Built an **OAuth 2.0** authentication portal for Spotify and Youtube, utilizing **Supabase** for persistent token caching and state management across multiple providers

Honck! - Productivity Assistant | *Groq, Python, React*

- Built a full-stack physical assistant with a **Raspberry Pi** and Bash scripts to enable continuous computer vision for monitoring and evaluating user focus, placing **top 30 best teams at Hack the North** out of 900+ participants
- Engineered a low-latency vision pipeline by integrating **Groq API** with custom system prompting to analyze video frames and generate real-time focus assessments
- Developed a **React Native** companion app to stream live camera feeds and display dynamic productivity metrics

ToyGFS | *C*

- Built a distributed file system in C over TCP sockets, modeled after the **2003 Google File System paper**
- Created a remote storage of large files with **0% data loss** by splitting files into multiple chunks across 3 servers
- Maximized system throughput by routing heavy file transfers directly between clients and storage nodes
- Automated file chunk recovery by detecting dead servers and re-replicating their chunks to maintain 3x redundancy